Estimating missing trait data in comparative biology using Bayesian Augmentation by Chained Equations: the BACE R package

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Abstract

1. Missing trait data are pervasive in comparative datasets limiting the types of questions that can be tested and compromising the power of comparative analyses. Standard approaches, such as complete-case analyses and single imputation, can bias inference and underestimate uncertainty, particularly when missing data is non-random. While robust imputation methods exits, many existing software are restricted to imputing only continuous or count variables and lack the capacity to impute discrete / categorical traits (e.g., multinomial, ordered categorical), which are common trait types in comparative datasets. Such limitations have prevented the widespread use of imputation approaches in comparative biology.
2. We introduce *BACE*, an R package implementing Bayesian Augmentation by Chained Equations for phylogenetic comparative data. *BACE* enables imputation with phylogeny-informed covariance that can account for both phylogenetic and non-phylogenetic species effects, along with mixed data types, in a fully Bayesian pipeline. *BACE* provides substantial flexibility in the imputation models fit, can decouple imputation models from analysis models, and propagates both parameter and imputation uncertainty in subsequent downstream analyses.

3. We show using simulations that *BACE* recovers missing trait values with a high degree of accuracy. Across 18 different scenarios that vary in sample size (n = 100, 300, 1000 species), phylogenetic signal (Pagel's $\lambda = 0.3, 0.7, 1$) and trait correlation ($\rho = 0, 0.5$), we show *BACE* recovers model parameters with little bias and nominal coverage. *BACE* outperformed other comparable frequentist approaches. We illustrate *BACE* using two case studies. The first imputes amphibian heat tolerance (continuous) under standardised conditions across 5,203 species using 2,661 experimental estimates from 524 species; 89.9% of species lack experimental measurements. The second uses AVONET data for 10,563 bird species to impute migratory behaviour (ordered categorical) and hand-wing index (continuous), addressing 17 naturally missing migration records (0.16%) and testing imputation after withholding approximately 20% of observed values for each trait.
4.

Introduction

Phylogenetic comparative methods (PCMs) provide a framework for understanding how traits evolve and covary across species (Felsenstein, 1985; Grafen, 1989; Martins & Hansen, 1997; Pagel, 1994). Expanding trait databases, increasingly comprehensive phylogenies, and greater access to environmental data are providing opportunities to investigate how evolutionary history and environmental conditions jointly shape variation and covariation among traits allowing a rich set of new evolutionary hypotheses to be tested (Blomberg & Todorov, 2025; Halliwell et al., 2025). However, missing trait data limit the questions that can be addressed, and excluding species with incomplete records can reduce sample sizes, statistical power and precision (Blomberg & Todorov, 2025; Little et al., 2024; Nakagawa & Freckleton, 2008). When the availability of trait data depends on species' characteristics, this exclusion can also bias estimates of trait relationships with significant consequences for inferences (Garamszegi & Møller, 2011; González-Suárez et al., 2012; Nakagawa & Freckleton, 2008). For example, incomplete trait coverage has been shown to alter inferred relationships between mammalian traits and extinction risk (González-Suárez et al., 2012). As such, there is a need for versatile and rigorous statistical approaches to deal with missing data in comparative biology.

Methods for handling missing data have a long history in statistics (Little & Rubin, 2002; Schafer & Graham, 2002). Approaches developed for comparative datasets include phylogenetic prediction of missing trait values (Bruggeman et al., 2009; Goolsby et al., 2017), as well as multiple-imputation procedures that account for phylogenetic and species-sampling uncertainty (Blomberg & Todorov, 2025; Pottier et al., 2025). Missing values can be accommodated within model fitting, for example through Bayesian data augmentation, which samples missing values alongside model parameters (Tanner & Wong, 1987). Alternatively, missing data can be handled in a multiple-imputation workflow, where m complete datasets are generated and analysed separately. Parameter estimates and their variances are then combined, commonly using Rubin's rules, to account for both within- and between-imputation uncertainty (Noble & Nakagawa, 2021; Rubin, 1987). Multiple imputations can be generated using a joint model for all variables or separate conditional models linked through chained equations. Chained equations cycle through these models, drawing missing values for each incomplete variable using the latest values of the others. This approach, also called fully conditional specification, allows different models for continuous and categorical variables and provides substantial flexibility in model structure (Buuren & Groothuis-Oudshoorn, 2011). Importantly, multiple imputation — like most missing-data methods — assumes data are missing at random

(MAR), and the plausibility of this assumption rests on imputation models that include the traits covarying with both the missing values and the probability of missingness (Gelman & Hill, 2007, 25; Noble & Nakagawa, 2021; Schafer & Graham, 2002). Because these relationships, and the types of the traits involved, differ from trait to trait, imputation requires substantial flexibility in the models fit to each one.

While a broad array of statistical methods and packages exist to deal with missing data, many are not suited to the datasets typical of comparative biology (Table 1). A few packages can use the phylogeny in their imputation models (e.g., *Rphylopars*, *phylomice*, *TDIP*), but most cannot (e.g., *mice*, *miceadds*, *jomo*, *pan*; Table 1). Moreover, the phylogeny-aware packages generally return a single completed dataset or single predicted values rather than multiple imputations, and most impute only continuous traits. A notable exception is *TDIP* (Gendre et al., 2024), which combines phylogenetic and machine-learning methods in an ensemble and can impute discrete traits. However, it too returns single completed datasets, and it cannot handle multi-level data structures, count data, or ordered categorical traits (Table 1). As such, there is a need for a statistically robust and flexible imputation approach that can handle multi-level data structures and a diversity of trait types, while ensuring imputation uncertainty is correctly accounted for in any downstream analyses and interpretations.

Here, we introduce a new Bayesian imputation software package for comparative biology, named *BACE* (**B**ayesian **A**ugmentation by **C**hained **E**quations), which uses the versatile Bayesian software *MCMCglmm* (Hadfield, 2010) to conduct multiple imputation through chained equations. *BACE* provides substantial flexibility in the imputation / analysis models fit, as well as a simple user interface, which makes it easy to apply in practice. *BACE* also provides options to decouple imputation and analysis models. Given its Bayesian approach, imputation and analysis models can be fit jointly and a final analysis model allows for pooling of posterior distributions to account for within and across imputation uncertainty. Additionally, samples from the posterior predictive distribution can be used to generate a set of m complete datasets and these datasets can be used with any modelling approach and pooled using Rubin's rules. We describe how *BACE* works, test its robustness using simulations, and demonstrate how to apply it using two case studies.

Table 1: R tools for imputing missing data, against the needs of phylogenetic comparative analysis. *Phylogeny* (Phylo.): a tree-derived covariance informs the imputation. *Proper multiple imputation* (MI): multiple imputed datasets (or an integrated Bayesian posterior) are produced, so downstream analyses reflect imputation uncertainty. *Multilevel* (Multi.): non-independence beyond the phylogeny (random effects / clustered rows). Remaining columns: an explicit imputation model for continuous (Cont.), count, binary (Bin.), ordinal (Ord., treated as ordered) and nominal (Nom., unordered multinomial) variables. ✓ = documented in the package's reference documentation (October 2026); (✓) = partial support; ✗ = not documented.

Package	Phylo.	MI	Multi.	Cont.	Count	Bin.	Ord.	Nom.	Reference
BACE	✓	✓	(✓)[a]	✓	✓	✓	✓	✓	this paper
Rphylopars	✓	✗	(✓)	✓	✗	✗	✗	✗	Goolsby et al. (2017)
phylomice[c]	✓	✓	✗	✓	✗	✗	✗	✗	Drhlik (2016)
TDIP[c]	✓	✗	✗	✓	✗	✓	✗	✓	Gendre et al. (2024)

Package	Phylo.	MI	Multi.	Cont.	Count	Bin.	Ord.	Nom.	Reference
BHPMF[c]	(✓)[b]	✗	(✓)	✓	✗	✗	✗	✗	Schrodt et al. (2015)
mice	✗	✓	(✓)	✓	✗	✓	✓	✓	Buuren & Groothuis-Oudshoorn (2011)
miceadds	✗	✓	✓	✓	✗	✓	(✓)[d]	✗[d]	Robitzsch & Grund (2026)
micemd	✗	✓	✓	✓	✓	✓	(✓)[d]	(✓)[d]	Audigier et al. (2018)
pan	✗	✓	✓	✓	✗	✗	✗	✗	Schafer & Yucel (2002)
jomo	✗	✓	✓	✓	✗	✓	(✓)[d]	✓	Carpenter & Kenward (2013)
mitml	✗	✓	✓	✓	✗	✓	✗	✓	Grund et al. (2023)
hmi[c]	✗	✓	✓	✓	✓	✓	✓	✓	Speidel et al. (2020)
mdmb	✗	✓	(✓)	✓	✗	✓	✓	✗	Robitzsch & Lüdtke (2024)
JointAI	✗	(✓)[e]	✓	✓	✓	✓	✓	✓	Erler et al. (2021)
smcfcs	✗	✓	✗	✓	✓	✓	✓	✓	Bartlett et al. (2015)
missForest	✗	✗	✗	✓	✗	✓	✗	✓	Stekhoven & Bühlmann (2012)

[a] Phylogenetic and species (replicated taxa) random effects; other grouping factors are not supported. [b] Taxonomic hierarchy (species/genus/family), not a tree with branch lengths. [c] Not on CRAN: *phylomice* and *TDIP* (GitHub only), *BHPMF* (archived 2017), *hmi* (removed 2022). [d] Ordinal/nominal support is limited: single-level methods only (*micemd*), predictive mean matching rather than an ordinal model (*miceadds*, whose multilevel methods also cannot impute nominal variables), or substantive-model-compatible outcome models only (*jomo*). [e] Imputation is integrated with the analysis model (fully Bayesian); imputed datasets can be exported.

The *BACE* R package: How it works?

BACE generates multiple completed datasets or pooled model(s) from comparative data with missing trait values (Figure 1). Users provide a data frame with species identifiers, a phylogeny with branch lengths, and model formulas specifying the predictors of each incomplete trait (Figure 1, step 1). Continuous, count, binary, ordered categorical and unordered categorical traits can be included in the same dataset. Each trait's type is declared by how its column is stored in the data

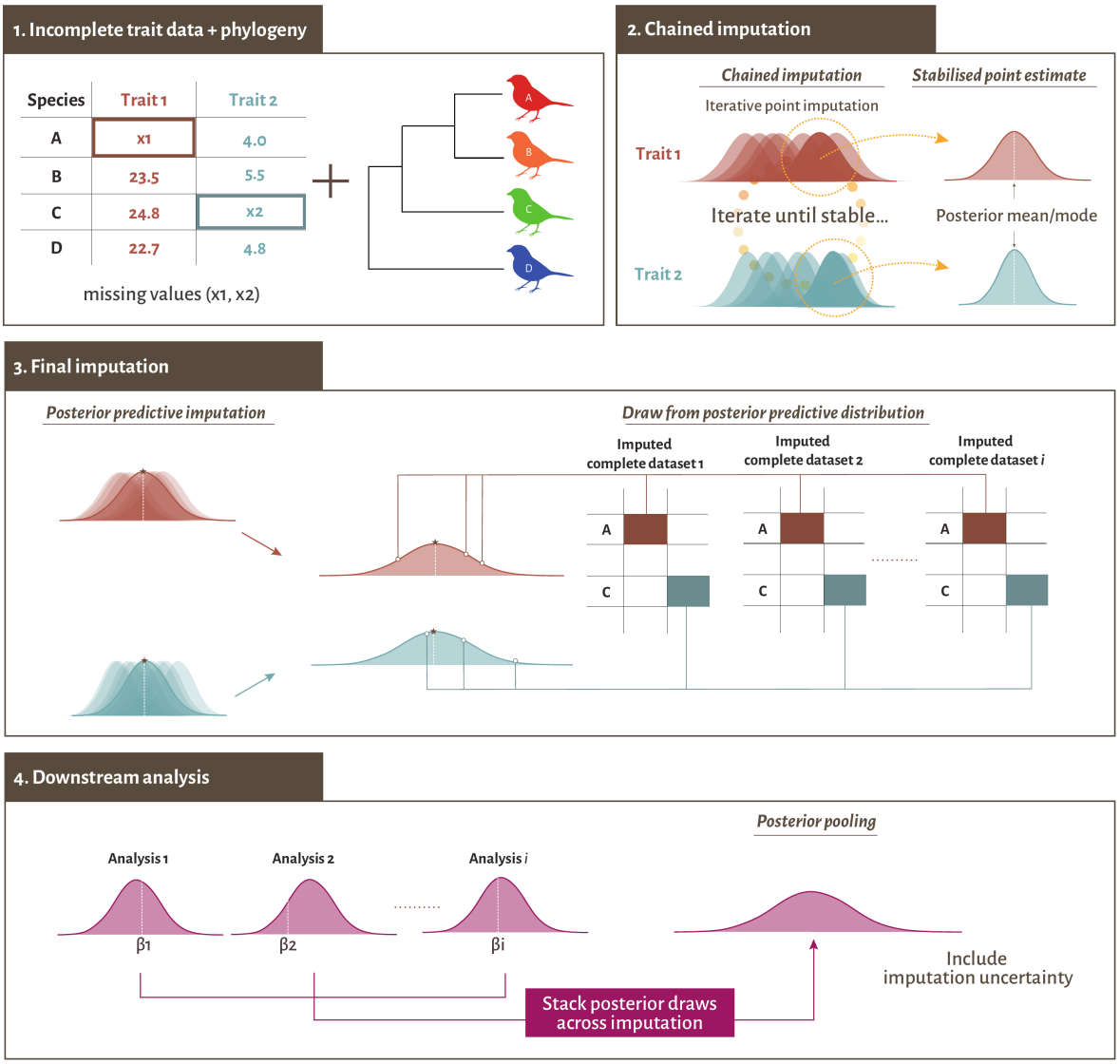


Figure 1: **Overview of the BACE imputation workflow**. (1) Inputs are an incomplete trait dataset and a phylogeny covering the species. (2) Chained imputation: a separate phylogenetic mixed model is fitted for each incomplete trait, and missing values are updated with point estimates until they stabilise. (3) Final imputation: models are refitted and missing values are drawn from their posterior predictive distributions. Repeated runs produce multiple completed datasets while retaining the observed values. (4) Downstream analysis: models are fitted to each completed dataset and posterior draws are pooled across imputations. Alternatively, parameter estimates and their variances can be combined using Rubin's rules to include imputation uncertainty.

frame (numeric for continuous, integer for counts, factors — ordered or unordered — for categorical traits). *BACE* then selects the correct error distribution for the model automatically. For example, if the response variable is a count a Poisson error distribution (log link) is used whereas if the response is a two-level categorical variable it will use a Bernoulli error distribution (logit link). Because the phylogenetic tree is user-supplied, evolutionary models other than Brownian motion can also be accommodated. For example, any model expressible as a branch-length transformation — Pagel's and , early burst, or single-optimum Ornstein–Uhlenbeck on an ultrametric tree — is fitted by transforming the tree before imputation, with the transform parameter estimated externally (Blomberg et al., 2003; Hansen, 1997; Pagel, 1999).

BACE fits a separate phylogenetic mixed model for each incomplete trait with *MCMCglmm* (Hadfield, 2010). Each model uses relationships among traits and shared evolutionary history to inform probable missing values. When species have repeated observations, a separate species effect can account for similarities not explained by the phylogeny (Cinar et al., 2022; Hadfield & Nakagawa, 2010; Nakagawa & Santos, 2012). Imputation has two phases. First, the models are fitted repeatedly, updating missing values with point estimates until these estimates stabilise (Figure 1, step 2). Once the chained equations stabilise, *BACE* cycles through the models again, drawing plausible missing values from their posterior predictive distributions (Figure 1, step 3). Repeating this process produces m completed datasets, each retaining the original observed values.

The main function `bace()` runs this workflow and returns completed datasets, fitted models and diagnostics. Posterior draws from the fitted imputation models can be pooled when their parameters are of direct interest. Pooled model parameters account for both parameter and between imputation uncertainty. Alternatively, users can fit a different analysis model to each completed dataset, combining posterior draws for Bayesian analyses or estimates and variances using Rubin's rules (Rubin, 1987; Zhou & Reiter, 2010). Either way, imputation uncertainty is carried into downstream analyses (Figure 1, step 4).

Typical *BACE* workflow

We demonstrate the typical user workflow in the code below, and Figure 2 maps the package's relevant functions onto the workflow steps, including the alternative Rubin's-rules pathway. A detailed tutorial with simulated data, and demonstrating our case studies, is also provided in the package documentation (https://daniel1noble.github.io/BACE/).

For users wishing to test out *BACE* or conduct simulations based on characteristics of their data, we also provide a simulation function `sim_bace()`. `sim_bace()` allows users to control the number of species, replicate observations per species, level of missing trait data, the distribution type of response and predictor variables, and the strength of the phylogenetic signal in each trait. Such simulations can help in planning analyses and identifying their power.

```
# 1) Prepare data and tree (rows = taxa; factors declared; counts as integers)
# 2) Run the full workflow: chained imputation -> convergence check ->
#    n_final posterior-predictive imputations -> pooled posteriors
res <- bace(fixformula     = "y ~ x1 + x2",
            ran_phylo_form = "~ 1 | Species",
            phylo          = tree,
            data           = dat,
```

```
            runs           = 10, n_final = 50)

# 3) Diagnostics
plot(assess_convergence(res$initial_results))

# 4a) Bayesian route: pooled MCMCglmm posteriors
summary(get_pooled_model(res, "y"))

# 4b) Rubin's-rules route: any downstream model per imputed dataset
fits <- with_imputations(res, function(d) lm(y ~ x1 + x2, data = d))
pool_mi(fits)
```

BACE users with existing species trait datasets can apply the *BACE* workflow by specifying the imputation and/or analysis model(s) in `bace()`. Model formulas can be provided as a single formula or a list of formulas, one for each missing response variable, defined by the argument `fixformula`. Users then supply the name of species-level random effect as a formula (using argument `ran_phylo_form`). If users have repeated species, then the random effect structure can be adjusted by setting the `species = TRUE` argument, which will add an additional non-phylogenetic random effect for species (Cinar et al., 2022; Hadfield & Nakagawa, 2010). If provided an analysis model with missing predictors, *BACE* will create equations for all variables with missing trait data automatically.

BACE provides a series of convergence diagnostics which help to identify possible issues during model fitting that need to be checked prior to inference. For example, within-chain MCMC convergence of each per-variable model as well as convergence of the chained-equations loop (the imputed-data sequence) are provided as output when `bace()` is run or which can be investigated after the imputation workflow is comeplete. `assess_convergence()` and `plot_*_convergence()` are two functions that provide users with various diagnositics on the change in imputed values as chained equations proceed. MCMC chains of any final model and parameter can be investigated using standard R functions (e.g., coda:). Such convergence information is impportant to identify when different priors or a different number of MCMC iterations are needed. Adjustments can be made in arguments of `bace()` to the priors used and teh number of MCMC iterations, burnin and thinning to ensure parameters are estimated with a sufficient effective sample size and that proper MCMC chain mixing is occuring.

Pooling pathways: stacked posteriors and Rubin's rules

(Structure only. Sources: `.agents/plans/pool_mi_rubin.md`; tutorial "Interpreting the pooled output"; Zhou & Reiter 2010; BDA3 Ch. 18.) - `pool_posteriors()`: stacking per-imputation chains approximates the marginal posterior; not Rubin's rules; O(1/M) difference at small M. - `with_imputations()` + `pool_mi()`: model-agnostic Rubin's rules (Barnard–Rubin df, fmi/riv); matches `mice::pool` to machine precision. - When to use which (variance components never via Rubin).

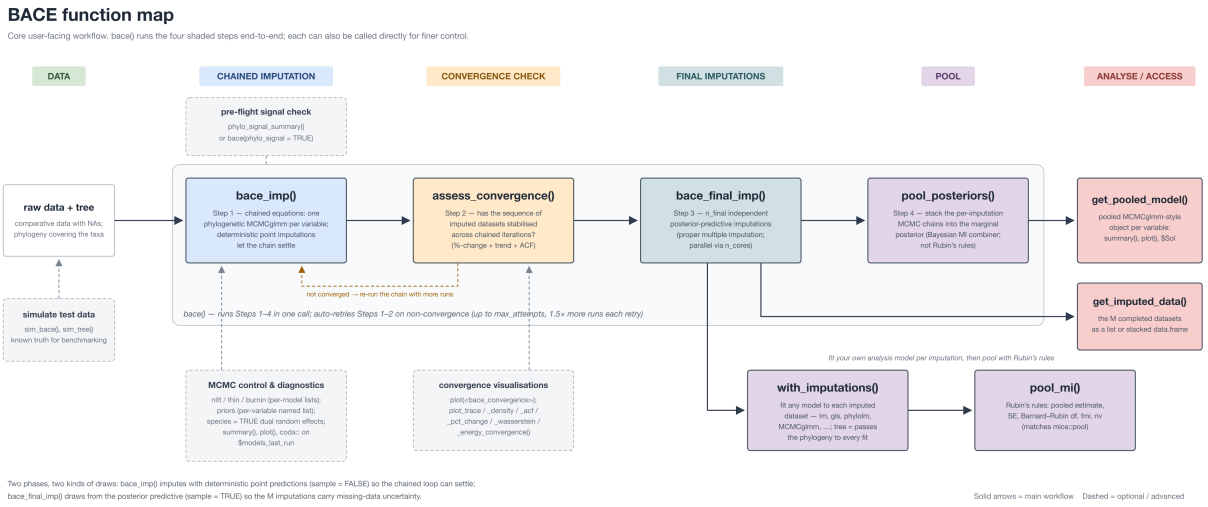


Figure 2: **Map of the core user-facing *BACE* functions**. Solid arrows show the main workflow from raw comparative data through chained imputation (`bace_imp()`), convergence assessment of the chained loop (`assess_convergence()`), posterior-predictive final imputations (`bace_final_imp()`) and posterior pooling (`pool_posteriors()`), with accessors (`get_pooled_model()`, `get_imputed_data()`) reading the finished object. The shaded band marks the steps run end-to-end by `bace()`, which automatically retries the chained phase on non-convergence. The lower-right branch is the alternative Rubin's-rules pathway: `with_imputations()` fits any downstream analysis model to each completed dataset and `pool_mi()` combines the estimates. Dashed boxes mark optional routes: simulating benchmark data (`sim_bace()`), pre-flight phylogenetic-signal screening (`phylo_signal_summary()`), per-model MCMC and prior control, and convergence visualisations.

Performance evaluation

Simulation design

(Structure only. Sources: `dev/12_recovery_simulation.R`, `dev/13_response_type_simulation.R`, `dev/simulation_results/SIMULATION_REPORT.qmd`; design details in `.agents/simulation-report.html`.)
- Study A: parameter recovery (gaussian slope) under MCAR/MAR; oracle and complete-case comparators. - Study B: five response types x MCAR/MAR; per-type accuracy/coverage metrics and their finite-draw/metric ceilings (`cover95_t`, prob-based sets, Bayes ceiling for categorical).

Parameter recovery and inferential calibration (Study A)

Imputation accuracy across response types (Study B)

Worked examples

We now turn to two case studies to demonstrate application of *BACE*. The first case study involves data from Pottier et al. (2025), whom imputed heat tolerance (continuous trait) under standardised conditions for 5,203 amphibian species of which only 524 (10.1%) have trait data. The second case study, uses BIRDBASE and AVONET to impute social (categocial and continuous data), for 10,563 bird species.

Discussion

(Structure only.) - What BACE adds relative to existing MI and phylogenetic-prediction tools. - Honest limitations - Practical guidance (pre-flight signal screening; when BACE is the wrong tool).

Availability and reproducibility

BACE (version 0.1.0) is free and open-source software released under the GPL (3). The source code and development version are available on GitHub (https://github.com/daniel1noble/BACE), where issues and feature requests can also be reported. A full tutorial — covering installation, when *BACE* is appropriate relative to other imputation tools, the complete workflow, and interpretation of pooled output and diagnostics — is available at https://daniel1noble.github.io/BACE/. A CRAN submission is in preparation; once accepted, the package will be installable directly via `install.packages("BACE")` (https://CRAN.R-project.org/package=BACE). Code reproducing all simulations reported here is included in the repository (`dev/`).

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(To write.)

Author contributions

Author contributions are reported using the CRediT taxonomy. **DWAN:** conceptualization, methodology, software, formal analysis, validation, visualization, writing — original draft, writing — review and editing, project administration. **SMD:** conceptualization, methodology, software, writing — review and editing. **AM:** software, validation, investigation (AVONET case study and tutorial), visualization, writing — review and editing. **PP:** data curation, investigation (amphibian heat-tolerance case study), writing — review and editing. **SN:** conceptualization, methodology, formal analysis, validation (independent simulation study), writing — review and editing. All authors approved the final version.

Data and code availability

(To write.)

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